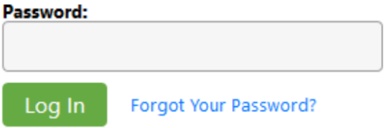
Session 4: PAKE-0, Assumptions, and Channels

Slides prepared by UMBC Protocol Analysis Lab

Last edited May 28, 2026

Passwords



- Can help prove your identity to somebody.
- We've used these forever.
- Should be kept secret and hard to guess.
- Often memorized...
- Don't store these in plain-text anywhere.

PAKE-0

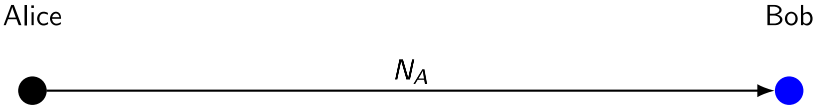
Alice

Bob

shared secret password: **pw**

● = Reception ● = Transmission

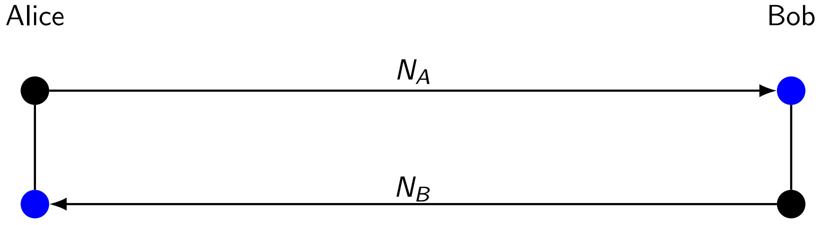
PAKE-0



shared secret password: **pw**

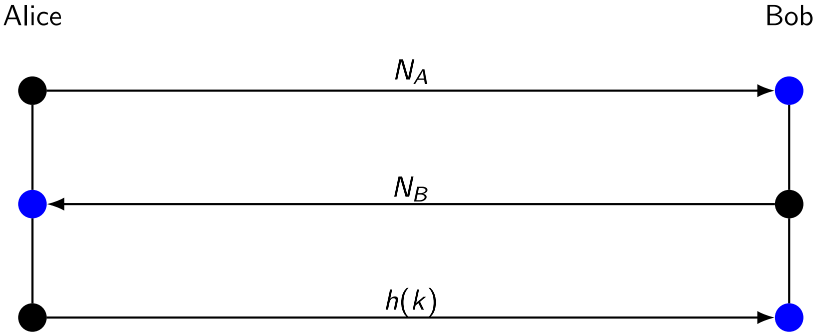
= Reception = Transmission

PAKE-0



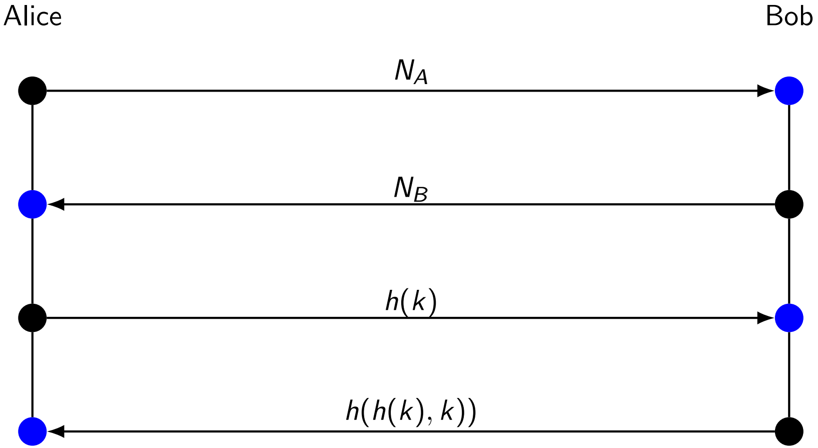
shared secret password: **pw**

PAKE-0



shared secret password: **pw** $\mathbf{k} = h(pw, Alice, Bob, N_A, N_B)$

PAKE-0



shared secret password: **pw** $\mathbf{k} = h(pw, Alice, Bob, N_A, N_B)$

= Reception = Transmission

PAKE-0 textbook diagram

shared secret password: pw

P		Q
$r \xleftarrow{R} \mathcal{R}$	$\xrightarrow{\quad r \quad}$	
		$s \xleftarrow{R} \mathcal{R}$
		$k \leftarrow H(pw, id_P, id_Q, r, s)$
$k \leftarrow H(pw, id_P, id_Q, r, s)$	$\xleftarrow{\quad s \quad}$	

session key: k

Figure 21.13: Protocol PAKE_0

Figure: Boneh, Dan, and Victor Shoup. "A graduate course in applied cryptography." Draft 0.5 (2020).

Assumptions in CPSA

Data origination assumptions:

- **uniq-orig** (e.g., nonce)
- **non-orig** (e.g., private key)
- **pen-non-orig** (e.g., pre-shared password)

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Now let's model it ourselves!

Homework

Homework

Model more in CPSA

Model more in CPSA

Now that you've seen PAKE modeled, can you model some more challenging protocol?
You will model a CTF exercise.

Survey

Fill in details about the survey here!